FULL STACK DEVELOPMENT PROJECT DOCUMENTATION

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Project Title	**Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer**
Team ID	SWTID-2026-1259
Date	30 September 2026

1. Introduction

• **Project Title:** Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

• **Team Members:** Team ID SWTID-2026-1259

2. Project Overview

• **Purpose:** The current IT procurement process lacks efficiency and automation, resulting in delays and manual overhead, particularly in handling standard laptop orders. Requests for standard laptops often require configuration, but this step is prone to oversight or delay, leading to frustration among users and inefficient resource allocation within the IT department. The objective of this project is to implement an automated workflow using ServiceNow Flow Designer to facilitate the procurement and configuration of standard laptops.

• **Key Features:**

  • Automated Catalog Task Generation upon Service Request approval.

  • Automatic task assignment to the Hardware team.

  • Seamless end-user experience reducing wait times.

  • Minimized manual intervention and procurement errors.

3. Architecture

• **Frontend / Service Catalog:** Built using ServiceNow Service Catalog interface. Users place requests for Standard Laptops (e.g., Lenovo Carbon x1) through configured Catalog Items with defined specifications.

• **Automation Engine (Backend):** Powered by ServiceNow Flow Designer running under 'System User' context in Global scope, utilizing 'Service Catalog' triggers and 'Create Catalog Task' actions.

• **Database Schema:** Leverages ServiceNow core tables including Catalog Item [sc_cat_item], Request [sc_request], Requested Item [sc_req_item], and Catalog Task [sc_task].

4. Setup Instructions

• **Prerequisites:** ServiceNow Instance with System Administrator privileges and Flow Designer plugin enabled.

• **Step-by-Step Installation:**

1. Navigate to All >> Flow Designer under Process Automation.

2. Create a new Flow named "Standard Laptop task" (Global Scope, Run As System User).
3. Add trigger: "Service Catalog".
4. Add action: "Create Catalog Task" with Short Description/Description set to "Laptop need to Configured", Assignment Group set to "Hardware", and Approval set to "Approved".
5. Save and Activate the Flow.
6. Navigate to Maintain Items >> "Standard Laptop" >> Process Engine tab >> Link "Standard Laptop task" flow and save.

5. Folder / Module Structure

• **Client / Service Catalog Module:** Service Catalog >> Hardware >> Standard Laptop Item Forms & Variables.

• **Process Automation Module:** Flow Designer >> Flows >> Standard Laptop task (Triggers, Actions, Data Pills).

• **Data Engine Module:** System Definition >> Maintain Items >> Process Engine & Core Tables (sc_request, sc_req_item, sc_task).

6. Running the Application

1. Open ServiceNow >> Service Catalog >> Hardware >> Standard Laptop.
2. Click "Order Now" to place request REQXXXXXXX.
3. Navigate to Request record >> Approvers >> Approve request.
4. Verify that the automated Flow generates a Catalog Task (SCTASKXXXXXXX) assigned to "Hardware" group.

7. API & Action Documentation

• **Action:** Create Catalog Task (ServiceNow Core Spoke)

• **Inputs:** Table [sc_task], Request Item [sc_req_item record], Short Description ["Laptop need to Configured"], Assignment Group ["Hardware"], Approval ["Approved"].

8. Authentication & Security

Workflow executes in "System User" context ensuring secure execution without depending on individual user session privileges, adhering to Role-Based Access Control (RBAC).

9. User Interface

Includes ServiceNow Service Portal Catalog forms, Flow Designer workflow builder canvas, Request Approval hierarchy tabs, and Catalog Task execution forms.

10. Testing

Validated via Flow Designer Test feature, end-to-end Service Catalog order submission, and User Acceptance Testing (UAT) verification of automated SCTASK generation.

11. Screenshots & Demo

Step-by-step visual documentation included in project submission media covering Flow properties, Catalog Task configuration, item binding, and generated task logs.

12. Known Issues

None. Workflow operates smoothly upon request approval.

13. Future Enhancements

Integration with automated hardware inventory asset tracking and user notification spokes.